NATNAEL G. MENGISTU

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Summary

AI/ML Engineer who builds production multimodal, agentic, and retrieval-grounded AI systems with safety guardrails, evaluation, and MLOps built in. Experienced in taking AI from proof of concept to deployment, grounding model outputs in source documents, and independently reviewing model exchanges for safety. Prior product owner of two telemedicine platforms built against clinical requirements. Combines hands-on Python and software-engineering practice with a responsible-AI mindset and clear communication of trade-offs to stakeholders.

Education

Penn State University

- Master of Artificial Intelligence — Expected 12/2026

Marist College, Poughkeepsie, NY

- Master of Science in Information Systems — 05/2022
- Bachelor of Science in Computer Science — 05/2019

Work Experience

AI/ML Engineer Intern, *Honeywell* — 06/2026 – 08/2026

- Owned end-to-end delivery of a multimodal extension to a production RAG support chatbot, letting users submit photos, video frames, and document pages instead of typing descriptions. Scoped the gap through stakeholder interviews across three teams, then designed, built, and deployed the solution.
- Extended the responsible-AI guardrail system to evaluate image and video inputs, preserving the safety coverage of the original text-only system. Tested against adversarial and edge case inputs.
- Architected a three-model pipeline: a vision-language model with OCR and image captioning that converts images, video, and documents into structured descriptions; an agentic reasoning model that iteratively calls existing RAG retrieval tools to ground answers; and an independent guard model that reviews every exchange in parallel, so safety review adds no extra response latency.
- Shipped three retrieval-grounded features: part identification from a photo, visible-damage flagging, and image-to-documentation search. Reduced hallucination risk by answering only from retrieved sources.
- Led agent-orchestration design for a multi-agent system that generated architectural and spatial designs collaboratively. Won the company intern hackathon among 8 teams.

Graduate Research Assistant, *Penn State University* — 08/2025 – Present

- Building a voice-controlled drone proof of concept that turns spoken commands into flight actions.
- Fine-tuning MotionGPT to recognize subtle human movements using reinforcement learning.
- Built and presented a proof-of-concept system combining computer vision and LLMs for interactive visual understanding, developed in collaboration with faculty as an educational and research tool.
- Developed real-time object detection and multi-object tracking pipelines in PyTorch and OpenCV.
- Implemented a natural language interface for text-based control and querying of vision models.

IT Specialist, *Global Disciples* — 09/2023 – 07/2025

- Strengthened data security by implementing encryption and access controls.
- Led the migration of 100+ user accounts from Google Workspace to Microsoft Exchange with zero downtime or data loss.
- Built an API integration between Salesforce and external platforms for real-time data synchronization.

- Provided technical support to Salesforce users across departments, improving issue resolution times by 30%.

Data Scientist & AI Consultant, *EnhanceIT* 02/2023 – 09/2023

- Architected ML pipelines on AWS using SageMaker, Lambda, and S3 for automated training and deployment.
- Designed RAG pipelines to ground chatbot and knowledge-retrieval systems in source documentation.
- Developed and fine-tuned LLM applications (GPT, BERT, LLaMA) for sentiment analysis, entity recognition, and summarization.
- Built and deployed computer vision models using TensorFlow and OpenCV for object detection and image classification.
- Built a real-time sentiment detection system tracking Reddit and Twitter activity to predict stock trends.

Senior Software Developer (ML & Fintech Focus), *AugArena* 09/2022 – 02/2023

- Developed a proof-of-concept AI-driven investment platform analyzing financial market data to generate predictive insights on stock movements.
- Integrated real-time data pipelines using FastAPI, PostgreSQL, and AWS for live market analytics.
- Designed ML models using sentiment analysis and macroeconomic indicators to forecast asset performance.

Product Owner, *MonitorMe* 03/2020 – 01/2021

- Led development and deployment of two telemedicine platforms, coordinating a distributed engineering team and prioritizing requirements against clinical and user needs.
- Owned release management and stakeholder communication to ensure on-schedule delivery.

Data Engineering Intern, *Galaxy.AI* 07/2019 – 09/2020

- Built web scraping pipelines with BeautifulSoup and ETL pipelines with Apache Spark, SQL, and Snowflake to prepare training data for ML models.
- Ran A/B tests on preprocessing techniques to improve model performance.

Technical Skills

- **Generative AI & ML:** LLMs, Retrieval-Augmented Generation (RAG), Agentic and Multi-Agent Systems, LLM Orchestration, LLM Fine-Tuning, Reinforcement Learning, Vision-Language Models, Computer Vision, NLP, DSPy, LLM Evaluation
- **Responsible AI & Security:** AI Safety and Guardrails, Grounded Retrieval, Data Security, Access Controls, Encryption
- **MLOps & Deployment:** MLflow, Docker, Kubernetes, CI/CD, AWS (SageMaker, Lambda, S3, Bedrock)
- **Frameworks:** PyTorch, TensorFlow, Scikit-learn, Hugging Face, OpenAI APIs, LangGraph, LangChain, OpenCV
- **Software Development:** Python, FastAPI, Django, Flask, REST API, GraphQL
- **Data Engineering:** Pandas, NumPy, SQL (PostgreSQL, MySQL), NoSQL (MongoDB, DynamoDB), Apache Spark, Hadoop, Redshift, Snowflake
- **Visualization:** Tableau, Power BI, Matplotlib, Seaborn